

Indian Institute of Technology, Gandhinagar
EE-226 Semiconductor Devices
Semester 2, 2025-2026
Tutorial-1

Q1. By what factor has the computation per square foot increased from ENIAC to the supercomputer? Interpretation?

**ENIAC was the first electronic,
Turing-complete device**



Area: 300 sq ft
Computation: 0.00289 megaflops

**Supercomputer,
Oak Ridge National Laboratory**



Area: ~7000 sq ft
Computation: ~1 exaflops

Q2. A facility has a power budget of 600 kW and wants to maximize its computational performance.

Given:

- ENIAC consumed 150 kW and had a performance of 0.00289 megaflops.
 - The Oak Ridge Supercomputer consumes 20 MW (i.e., 20,000 kW) and delivers 1 exaflop (i.e., 1×10^6 megaflops)
- a) How many ENIACs can be powered with the available 600 kW, and what would be the total computational performance in megaflops?
- b) What fraction of a single Oak Ridge supercomputer could be powered with 600 kW, and what would be the corresponding computational performance in megaflops?
- c) Based on power efficiency (megaflops per kW), which system is more efficient?

Q3. Write down the entities that you know of which fit in different powers of 10

a. E.g. in the range of

$< 10^{-15}, 10^{-12}, 10^{-9}, 10^{-6}, 10^{-3}, 10^0, 10^3, 10^6, 10^9, 10^{12}, 10^{15}$ etc

b. Plot x vs e^x with x varying from 1 to 20.

i. In linear scale with y-axis from 1 to 10, 1000 to 10000 etc. Can we visualize all the values?

ii. Plot the same in log scale. Can we visualize all the values? What else can we visualize.

c. If you plot a set of data points on a log-linear plot (x-axis linear, y-axis logarithmic) and the points fall on a straight line, what kind of mathematical behaviour does this suggest?

Q4. How slow is the fastest manmade object

a. Compared to speed of light?

b. Compared to maximum electron velocity in vacuum?

If a device is length is $1\mu m$, assuming electron achieved its maximum velocity, how much time is required to cross $1\mu m$? Compare this with the latest processor speed. Do we have any scope for improvement – any suggestions?

Q5. Typically, billions transistor are placed in a processor chip of a laptop today. Typically, how much area would be required for one transistor assuming 10 billion transistor on 5 cm x 5 cm chip?

Q6. Read the attached articles

1. Plenty of Room at the Bottom

2. A company of legend: The legacy of Fairchild Semiconductor

a. The article “Plenty of Room at the Bottom” is dated Dec 1959 i.e. more than 60 years before.

i. What is the core message from the article?

ii. What are the methods suggested to write on the tiny (few atoms) scale?

iii. What are the methods suggested to read such tiny scales?

iv. What did Alber R. Hibbs suggest to Richard Feynman? Do we see such machines existing today?

- b. In the paper “A Company of Legend-The Legacy of Fairchild Semiconductor” and answer the following questions:
- What packaging innovation by Fairchild helped standardize IC integration onto PCBs, and why was it important?
 - What was the importance of the planar process in enabling the integrated circuit revolution?

Q7. If a metal atom in a crystal lattice is surrounded by eight nearest neighbor atoms, what is its coordination number? Explain why the coordination number is not always equal to the number of bonds the atom forms in a molecule?

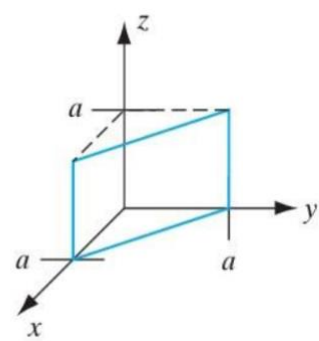
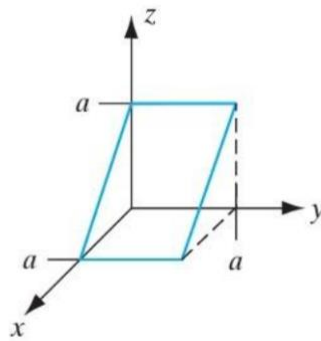
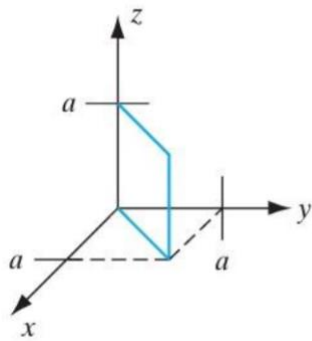
Q8. A crystal structure has lattice constant $2.4 \text{ \AA}$. The atomic radius is $1.04 \text{ \AA}$. What is the possible lattice structure: hcp/bcc/fcc?

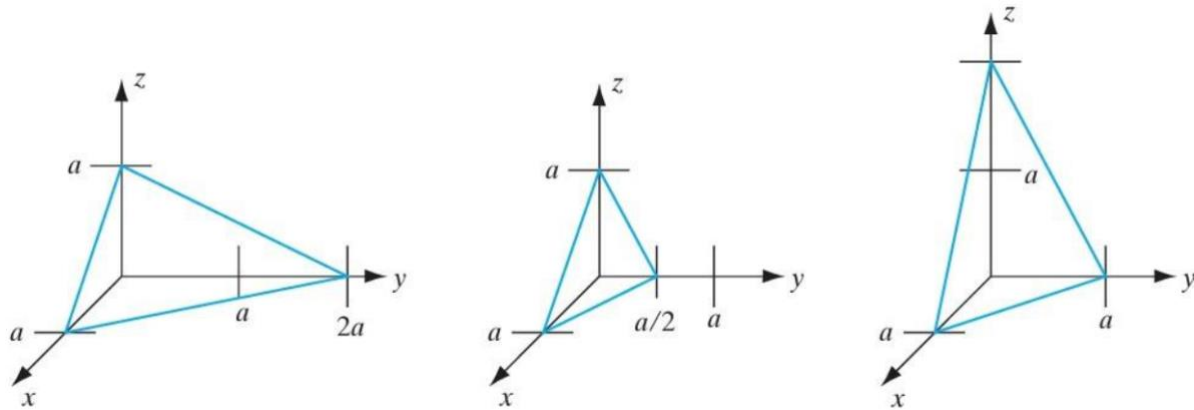
Q9. Assuming a Simple Cubic crystal structure, make a sketch of the following planes.

- a) (100) b) (123) c) (111) d) (221) e) (110) f) $(1'10)$ g) $(1'1'1')$

NOTE: $1' = 1$ bar (If 1 bar encountered in any problem you must consider negative values of any coordinate)

Q10. Find the Miller indices of the planes given below.





Q11. For Simple Cubic (SC), Body Centered Cubic (BCC) and Face Centered Cubic (FCC) structure:

- Find the number of atoms
- How far apart in terms of lattice constant are the nearest neighbour atoms in each case, measured from center to center
- Find atomic packing fraction (APF) if the atoms are closely packed together
- Find volume density ($\#/cm^3$) if lattice constant $a = 5 \text{ \AA}$
- Calculate the areal density of atoms ($\#/cm^2$) on the (100) plane

Q12. For a bcc lattice structure with a lattice constant of $3 \text{ \AA}$,

- Calculate the separating distance between the nearest atoms.
- The radius and the volume of each atom.

Q13. For Silicon (Diamond crystal structure) with the lattice constant of $5.43 \text{ \AA}$.

- How to get diamond crystal structure from FCC?
- Is diamond structure a Bravais Lattice? Why?
- Calculate APF.
- Find volume density ($\#/cm^3$)
- Calculate the areal density of atoms ($\#/cm^2$) on the (100) and (111) plane.

Q14. Do GaAs and Si have identical crystal structure? Explain.

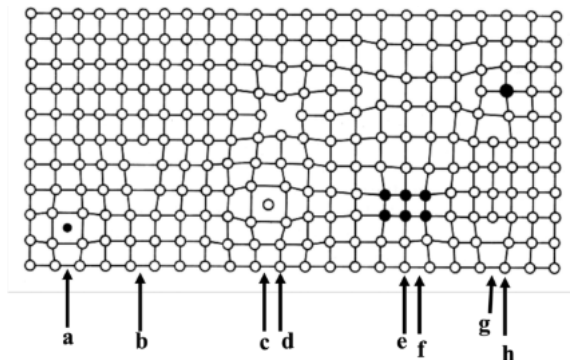
Q15. What difference can we spot in the conductivity values of semiconductors, conductors and insulators? Explain with the case of Si, Ag and Diamond?

Q16. Based on the observation in Q2, explain why semiconductors are preferred over conductors and insulators in electronic applications.

Q17. What is a unit cell? Can there be more than 1 unit cell in the crystal structure? Why?

Q18. What is a primitive cell? Can there be more than 1 primitive cell in the crystal structure? Why?

Q19. Identify the defects present in the given figure. Are any defects useful?



Q20. Describe the arrangement and movement of particles in solids, liquids, and gases?

Q21. If you have a 300 mm silicon wafer, how many numbers of chips can you produce approximately from a single wafer, if the size of each chip is 1mm x 1mm. Is it a straightforward division?

Q22. If the dimensions and power consumption of a vacuum tube for a single bit are 5cm x 6cm x 9cm and 50W respectively, what would be the total volume of vacuum tubes required, and the total power consumed by the vacuum tubes for 1 GB of information (Gigabytes). Relate the volume to the existing monuments and power consumption to nuclear power plants capacity. Calculate approximately.

Q23. What are the three broad categories of companies in the Semiconductor ecosystem with respect to design and manufacturing. What are the functionalities of each category. Name any two companies from each category

Q24. What is the difference between a lattice and a crystal? How many different one-dimensional lattices can you have?

Q25. Explain the importance of Atomic Packing Factor (APF) and its impact on material properties.

Q26. Why areal density or surface density is important? Is current dependent on this – how?